Enhancement One Narrative

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CS-499: Computer Science Capstone

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September 20, 2026

1. Briefly describe the artifact. What is it? When was it created?

   The artifact is my InputOutput.cpp file, which is the location of all the input and output for my Investment Calculator application. I initially created this for my CS-210 Programing Languages, back in February 2025.

2. Justify the inclusion of the artifact in your ePortfolio. Why did you select this item? What specific components of the artifact showcase your skills and abilities in software development? How was the artifact improved?

   I selected this artifact because it was one of my first C++ projects and I definitely did not have strong skills in software engineering and design. There were a few areas where I did not make it clear enough to the user on what was expected or even what the program did. The components of my artifact that showcase my skills and abilities in software development is how the program itself flows, with clear instructions and intent given to the user. I felt that I was assuming the user knows what the program wants from them in terms of user input.

   I improved this artifact by adding an additional method called printWelcomeScreen(), which just prints a small welcome message to the screen with a small description on what the program does. I also went through all the methods within this artifact and made it clear in the output statements what is acceptable input for each variable. I also cleaned up some inline comments and put a header comment along with comment blocks above each method.

3. Did you meet the course outcomes you planned to meet with this enhancement in Module One? Do you have any updates to your outcome-coverage plans?

   I felt that I have met the course outcomes that I planned to meet with this enhancement, which are:

   - Employ strategies for building collaborative environments that enable diverse audiences to support organizational decision making in the field of computer science.
   - Design, develop, and deliver professional-quality oral, written, and visual communications that are coherent, technically sound, and appropriately adapted to specific audiences and contexts.
   - Demonstrate an ability to use well-founded and innovative techniques, skills, and tools in computing practices for the purpose of implementing computer solutions that deliver value and accomplish industry-specific goals.

4. Reflect on the process of enhancing and modifying the artifact. What did you learn as you were creating it and improving it? What challenges did you face?

   Looking back on my process of enhancing and modifying this artifact, I learned through discovery that there was by far more user clarification than I first suspected. Once I got into the groove of enhancing and testing, I found that there could possibly be more user confusion in how the format for input was expected. I then discovered that the Investment Calculator application was not protected from users entering extreme values, for example when the user is asked for the length in years that they want to calculate the compounding interest, they were able to input how ever many years that they want. This could not only break the application or cause unintended consequences or outcomes, due to the possibility of inputting some extremely large value like 100,000 years. This is planned to be fixed with enhancement two, because it has to do with algorithms and data structures. I have a tendency to do a thing called feature creep, which is where I do more than what is required in terms of coding. This could be a bad thing when working in teams where we all have our predefined requirements, by adding additional features could break or cause confusion.

   The biggest challenge that I faced during the first enhancement was deciding on how I should go about modifying and enhancing the design, there is always more than one way to do something. I decided to stick with the least complex fix, which should allow for easy modification/updating in the future. The next challenge that I faced and continue to face with all my projects is in how I write my comment blocks above each code block. I tend to explain what each line does rather than what the block of code does as a whole.

   To overcome these challenges, I keep a sticky note attached to my computer monitor with the words: “feature creep”, and “comments = what it does as a block of code“. I do understand that sometimes inline comments are needed but only to explain a line of code that is not clear or that is unusual.